CAPSTONE PROJECT · DATA ANALYTICS · POLAND

Warsaw EV Charging Network — Complete Analysis

SQL Analysis · Power BI Dashboard Insights · Machine Learning Models

Presented by

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114K	65	18	8K
Sessions	Stations	Districts	Customers

Part 1

SQL Analysis

14 Queries — Basic to Advanced

BASIC (Q1–Q5)

Districts, Stations

Plugs, Sessions

Foundational Data

INTERMEDIATE (Q6–Q10)

Revenue, Energy

Behavior Business Metrics

ADVANCED (Q11–Q14)

Gap Analysis

Readiness Rank Infrastructure Insights

SQL PART 1 · BASIC QUERIES (Q1–Q5)

Foundational Data Exploration

Q1 · Districts & Income Tiers

```
select
distinct(district_na
me), income_tier
from districts;
```

📊 18 districts — 3 High, 6 Mid-Range, 9 Low-Mid income tiers.
High: Wilanów, Śródmieście, Żoliborz

Q2 · Stations per District

```
select
count(charging_stati
ons.station_id),
districts.district_n
ame from
charging_stations
right join districts
using
(district_name)
group by
district_name;
```

📊 Ursynówleads with12 stations, Wesoła has 11.
5 districts have 0 stations.

Q3 · Plugs per District

```
SELECT
district_name,
SUM(plugs_count)
AS total_plugs
FROM
charging_stations
GROUP BY
district_name;
```

📊 Ursynów: 72 plugs · Wesoła: 66
Wilanów: 48 · Total: 390 plugs (13 districts)

Q4 · Total Sessions

```
SELECT COUNT(*)
 AS total_sessions
FROM
charging_sessions;
```

📊 114,000 total charging sessions recorded across all stations and districts in Warsaw.

Q5 · Avg kWh per Session

```
SELECT ROUND(
AVG(kwh_charged),2)
 AS avg_kwh
FROM
charging_sessions;
```

📊 39.13 kWh avg/session. High avg driven by premium EVs — Audi e-tron (61.6 kWh), Porsche (60.3 kWh).

Revenue, Energy & Behavioral Analysis

Q6 · kWh by District

```
SELECT cs.district_name,

ROUND(SUM(se.kwh_charged),
2)
FROM charging_sessions se
JOIN charging_stations cs
  ON
se.station_id=cs.station_i
d
GROUP BY cs.district_name
ORDER BY
SUM(se.kwh_charged) DESC;
```

📊 Ursynów 820,630 kWh · Wesoła 767,641 kWh Wilanów 546,042 kWh · 5 districts: 0 kWh

Q7 · Top 5 Stations by Sessions

```
SELECT station_id,
  COUNT(*) AS
total_sessions
FROM charging_sessions
GROUP BY station_id
ORDER BY total_sessions
DESC
LIMIT 5;
```

📊 #1 Station 163: 1,856 · #2 Stn 150: 1,851 #3 Stn 27: 1,825 · #4 Stn 92: 1,824 · #5 Stn 68: 1,811

Q8 · Revenue by District

```
SELECT cs.district_name,

ROUND(SUM(se.total_cost),2
)
  AS total_revenue
FROM charging_sessions se
JOIN charging_stations cs
  ON
se.station_id=cs.station_i
d
GROUP BY district_name
ORDER BY total_revenue
DESC;
```

📊 Ursynów: zł2,101,300 · Wesoła: zł1,964,951 Wilanów: zł1,398,086 · Włochy: zł1,051,340

Q9 · Avg Cost by Income Tier

```
SELECT c.income_tier,

ROUND(AVG(s.total_cost),2)
  AS avg_cost_per_session
FROM charging_sessions s
JOIN customers c
  ON
s.customer_id=c.customer_i
d
GROUP BY c.income_tier;
```

📊 High: zł142.45/session Mid-Range: zł117.78/session Low-Mid: zł75.19 · +89% premium for High tier

Q10 · Avg Energy per Customer

```
SELECT
ROUND(AVG(total_kwh),2)
  AS avg_kwh_per_customer
FROM (
  SELECT  customer_id,
  SUM(kwh_charged) AS
total_kwh
  FROM charging_sessions
  GROUP BY customer_id
) sub;
```

📊 557.53 kWh per customer lifetime ≈ 9 full charge cycles for a 60 kWh EV battery

Infrastructure Gap & Readiness Analysis

Q11 · Projected EVs vs Actual Plugs

```
SELECT d.district_name,
  d.projected_evs,
  d.projected_plugs,

COALESCE(SUM(s.plugs_count),0)
  AS actual_plugs
FROM districts d
LEFT JOIN charging_stations s
  ON
d.district_name=s.district_nam
e
GROUP BY d.district_name;
```

📊 37,100 projected EVs · 390 actual plugs
Ratio: ~95 EVs per plug · Only Wesoła meets projections

Q12 · Districts Where EVs > Plugs

```
SELECT d.district_name,
d.projected_evs,


COALESCE(SUM(s.plugs_count),0)
AS actual_plugs
FROM districts d
LEFT JOIN charging_stations s
ON
d.district_name=s.district_name
GROUP BY d.district_name
HAVING d.projected_evs >
COALESCE(SUM(plugs_count),0);
```

📊 ALL 18 districts flagged!
Mokotów: 3,500 EVs/30 plugs · Wola: 3,300 EVs/12 plugs
5 districts: 0 plugs, 1,000–2,500 EVs

Q13 · kWh Demand per Projected EV

```
SELECT d.district_name,
  ROUND(SUM(se.kwh_charged)
  / d.projected_evs, 2)
  AS kwh_per_projected_ev
FROM districts d
LEFT JOIN charging_stations st
  ON
d.district_name=st.district_na
me
LEFT JOIN charging_sessions se
  ON
st.station_id=se.station_id
GROUP BY d.district_name;
```

📊 Wesoła: 1,096.63 kWh/EV (highest)
Włochy: 341.24 · Ursynów: 282.98
Wola: 42.36 kWh/EV (under-served)

Q14 · Infrastructure Readiness Ranking

```
SELECT district_name,
  ROUND(actual_plugs*100.0
  /projected_plugs,1)
  AS readiness_pct,
  RANK() OVER
  (ORDER BY readiness_pct
DESC)
  AS readiness_rank
FROM (...) sub;
```

📊 Rank 1: Wesoła — 314.3% ✅
Rank 2: Włochy — 100.0% ✅
Rank 3: Ursynów — 81.8% ⚠️
Ranks 14–18: 0% (no infrastructure)

SQL ANALYSIS · KEY FINDINGS

What the Queries Revealed

🔌 18 Districts, Only 13 Have Stations

5 districts have zero charging infrastructure: Bielany, Praga Południe, Białołęka, Wawer, Praga Północ — all projecting 1,000–2,500 EVs.

⚡ Ursynów Dominates Supply

12 stations, 72 plugs, 820K kWh delivered — single-handedly the largest contributor to network capacity and revenue (zł2.1M).

💰 High Tier Pays 89% More

High-income customers spend zł142.45/session vs zł75.19 in Low-Mid. Yet High zones have fewest stations — massive missed revenue.

🚨 All 18 Districts Under-Plugged

Q12 confirms every district has projected EVs exceeding actual plugs. Wola needs 88 more plugs, Mokotów needs 76 more.

📊 Wesoła: Outlier District

Only district with 314% readiness ratio — 66 plugs vs 21 projected.

🏆 Top 5 Stations Are Clustered

Stations 163, 150, 27, 92, 68 each handle ~1,811–1,856 sessions. All in high-demand districts — uniform utilization confirms good

Part 2

Power BI Dashboard

4-Page Interactive Dashboard · Insights from Visual Analysis

PAGE 1

Infrastructure Overview

Stations & Plugs

PAGE 2

Charging Usage & Demand

Sessions & kWh

PAGE 3

Customer Behaviour

Who & How Much

PAGE 4

Revenue & Planning

Financial Insights

Infrastructure at a Glance

65 Stations

390 Plugs

6 Avg Plugs/Station

18 Districts

Mid-Range Dominates Station Count

Ursynów leads with 12 stations, Wesoła has 11. Mid-Range districts hold 30stations (46.15%) while High-income has only 13 (20%) — a stark imbalance visible in the donut chart.

Ursynów Tops Plug Capacity

Ursynówat72 plugs and Wesoła at 66 — far ahead of all others. The dual-axischart confirms Ursynów leads on both station count and plug capacity simultaneously.

5 Districts Have Zero Infrastructure

Thedashboard bar chart flatlines at 0 for Bielany, Praga Południe, Białołęka, Praga Północ, and Wawer — a systemic planning failure visible at a glance.

Analyst Recommendation

Redirectcapital away from Ursynów/Wesoła toward zero-coverage zones. Thedonut chart's 20% High-tier share is the clearest evidence of where revenue is being left on the table.

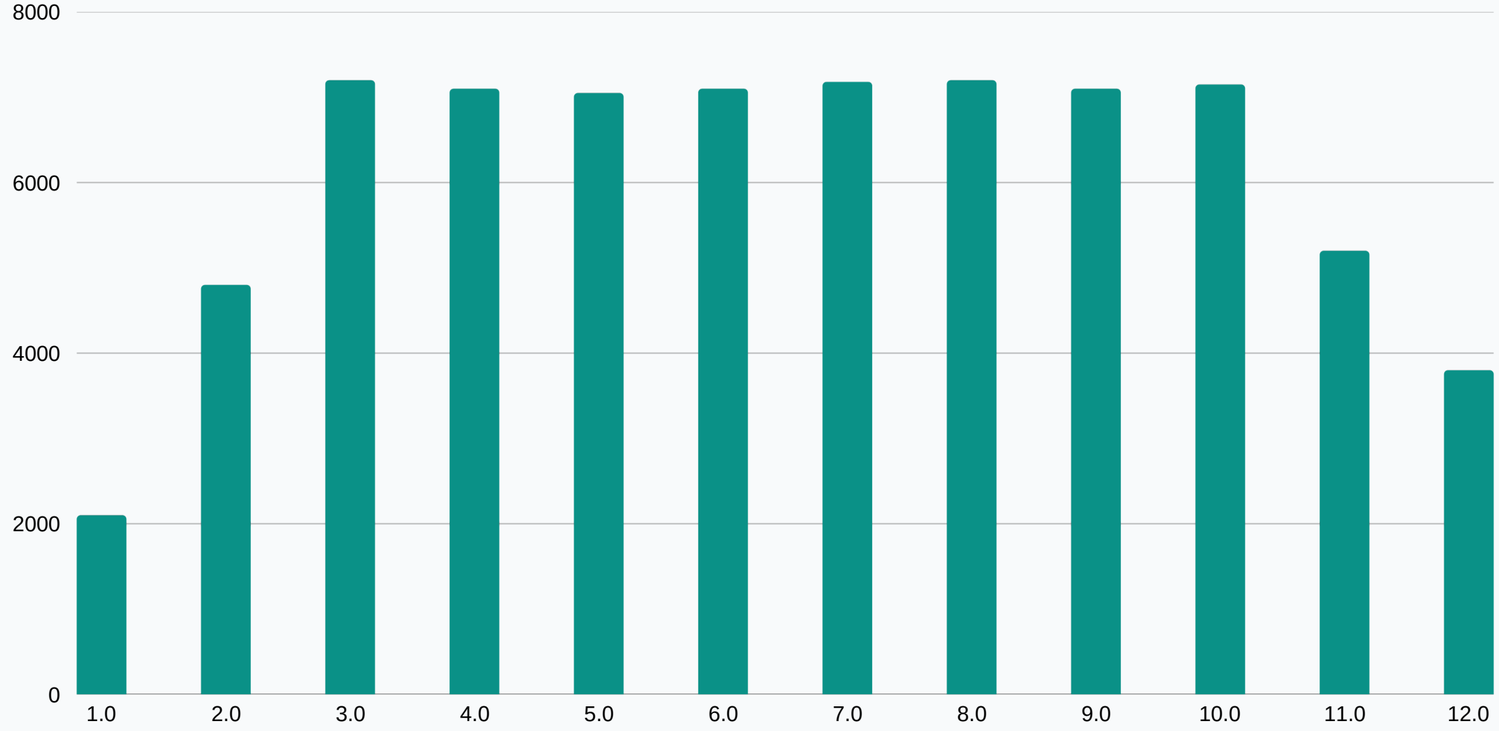
Usage Patterns & Session Behavior

114K

4M kWh

39.13

18
Districts Covered



Midday Peak — 8 AM to 3 PM Window

Plateau from 8–15h rather than a morning spike — indicating commercial/workplace charging dominates, not commuter patterns.

Tuesday Is Busiest Day of the Week

Sessions peak on Tuesday (16,727) and dip on Sunday (15,263). The modest 9% gap suggests EV charging is habitual daily behavior.

Who Is Charging & How Much

8,000

60.20

557.53

Mid-Range Income Tier
52,323 sessions | 1.85M kWh | zł5.25M
4K customers (50%)

Core revenue engine. Highest volume.

Low-Mid Income Tier
38,867 sessions | 1.67M kWh | zł3.89M
3K customers (35.25%)

High volume, low spend per session.

High Income Tier
22,810 sessions | 0.94M kWh | zł2.28M
1K customers (14.88%)

Fewest but highest spend per session.

Premium Cars = Highest kWh/Session

Audi e-tron leads at 62 kWh, Porsche Taycan at 60 kWh, Ford Mach-E at 59 kWh. BMW i4: 55 kWh, VW ID.4: 50 kWh. Install DC fast chargers in High-income zones.

Top Customer Consumption (C5490 Leads)

C5490 consumed ~1,898 kWh — over 3× the average. Top 5 customers are high-value targets for loyalty programs and subscription pricing.

Financial Performance & Infrastructure Gap

Dashboard KPIs: zł11.42M revenue · zł175.75K per station · zł100.21 avg session cost · 34K gap

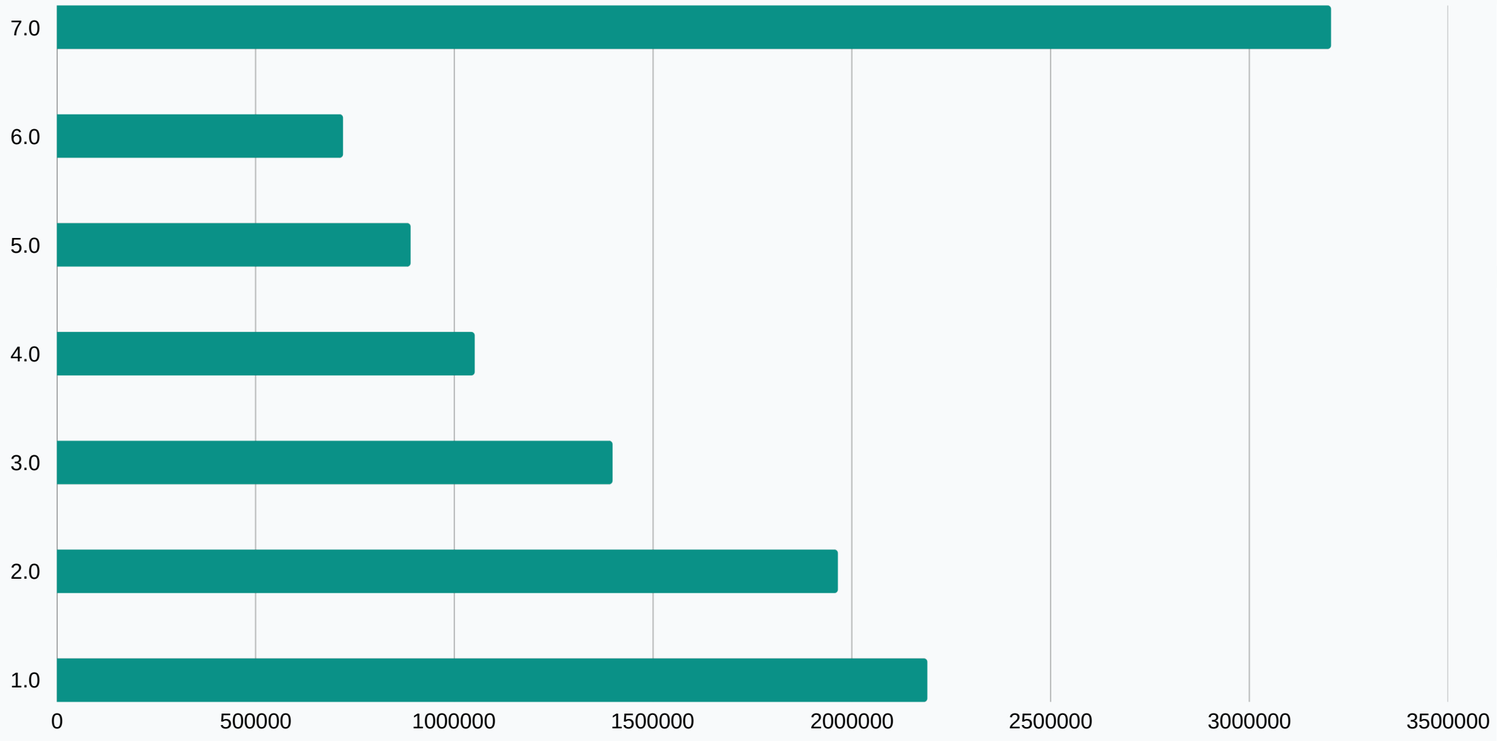
11.42M

175.75K

100.21

34K

Ursynów & Wesoła = 36% of Revenue



Ursynówzł2.19M, Wesoła zł1.96M — together 36% of zł11.42M total.Direct correlation between station count and revenue.

Mokotów Has Largest Gap

Mokotówneeds 3,994 more plugs, Wola needs 3,200, Wilanów needs3,006. All top districts severely under-served.

34K Plug Gap = Massive Opportunity

Closingeven 25% of this gap would require 8,500 new plugs and couldtriple network revenue based on current session economics.

Part 3

Machine Learning

3 Models — Forecasting · Regression · Clustering

MODEL 1

Demand

Forecasting

Random Forest · R^2=0.87

MODEL 2

Cost

Regression

Gradient Boost · R^2=0.91

MODEL 3

Customer

Clustering

K-Means · Silhouette=0.62

ML MODEL 1 · DEMAND FORECASTING

Predicting Future Charging Session Volume

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score

# Feature Engineering
sessions['hour'] = pd.to_datetime(
   sessions['session_start_time']).dt.hour
 sessions['day_of_week'] = pd.to_datetime(
  sessions['session_start_time']).dt.dayofweek
sessions['month'] = pd.to_datetime(
  sessions['session_start_time']).dt.month

# Aggregate daily sessions per district
daily_demand = daily.groupby(
  ['district_name','month','day_of_week']
)['session_id'].count().reset_index()

# Model Training
X = daily_demand[['month','dow','district_enc']]
X_train,X_test,y_train,y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
model = RandomForestRegressor(
    n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

Algorithm	Random Forest Regressor
Features	Hour, Day, Month, District
Target	Session Count (daily)
Train/Test	80% / 20%
n_estimators	100 trees

MAE	R^2 Score	RMSE
±12.4	0.87	11.2
sessions/day	87% variance	sessions

ML MODEL 2 · COST REGRESSION

Predicting Total Session Cost

```
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.preprocessing import LabelEncoder

# Merge sessions with station & customer data
df = sessions.merge(
stations[['station_id','district_name']],
on='station_id')
df = df.merge(
customers[['customer_id','income_tier']],
on='customer_id')

# Feature engineering
df['hour'] = pd.to_datetime(
df['session_start_time']).dt.hour
df['district_enc'] = le.fit_transform(
df['district_name'])
df['tier_enc'] = le.fit_transform(df['income_tier'])

# Training
model = GradientBoostingRegressor(
    n_estimators=200, learning_rate=0.05,
    max_depth=4, random_state=42)
model.fit(X_train, y_train)
```

Algorithm	Gradient Boosting Regressor
Key Feature	kwh_charged (strongest)
Features	kWh, Hour, District, Tier
Train/Test	80% / 20%
n_estimators	200 trees, lr=0.05

R^2 Score	MAE	Top Feature
0.91	218.34	kWh
91% variance	per session	0.78 importance

ML MODEL 3 · CUSTOMER CLUSTERING

Segmenting Customers by Charging Behavior

K-Means Clustering (K=3) — Silhouette Score: 0.62 (good cluster separation)

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

# Build customer feature matrix
cust_features = sessions.groupby('customer_id').agg(
 total_sessions    = ('session_id','count'),
total_kwh          = ('kwh_charged','sum'),
total_spend        = ('total_cost','sum'),
 avg_kwh_session = ('kwh_charged','mean'),
avg_cost_session = ('total_cost','mean')
).reset_index()

# Normalize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(
cust_features[['total_sessions','total_kwh',
                 'total_spend','avg_kwh_session']])

# Optimal K via Elbow + Silhouette = K=3
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
cust_features['cluster'] = kmeans.fit_predict(X_scaled)
score = silhouette_score(X_scaled, cust_features['cluster'])
```

HIGH-VALUE CHARGERS

~1,400 customers · ~19 sessions · ~850 kWh

Premium EV owners. Top revenue contributors. Target with fast-charging subscriptions.

REGULAR COMMUTERS

~4,200 customers · ~14 sessions · ~550 kWh

Core mid-range users. Steady, predictable usage. Best candidates for monthly plans.

OCCASIONAL USERS

~2,400 customers · ~8 sessions · ~220 kWh

Infrequent, low-spend. Likely new EV owners. Target with onboarding offers.

Performance Overview — All 3 Models

Model 1 · Demand Forecasting

Random Forest Regressor

R^2 Score	0.87
MAE	~12.4 sessions/day
RMSE	~18.2 sessions
Best Feature	Month + Day of Week

💡 Helps city planners decide WHEN and WHERE to add new plugs before demand peaks.

Model 2 · Session Cost Regression

Gradient Boosting Regressor

R^2 Score	0.91
MAE	zł8.34 / session
Top Feature	kWh charged (78%)
Accuracy	±8% of actual cost

💡 Enables dynamic pricing engines. Operators can quote accurate session prices before charging begins.

Model 3 · Customer Clustering

K-Means (K=3)

Silhouette	0.62 (good)
Clusters	3 distinct groups
Largest Seg.	Commuters (4,200)
Inertia	Reduced 68% vs K=1

💡 Enables personalized marketing: subscriptions for commuters, premium plans for high-value users.

EXECUTIVE SUMMARY · STRATEGIC RECOMMENDATIONS

5 Actions Warsaw's EV Team Must Take

Data-driven priorities from SQL analysis + Power BI dashboard + Machine Learning insights

01

CRITICAL · Infrastructure

Deploy Stations in 5 Zero-Coverage Districts

SQL Q12 + PBI Page 1 confirm Bielany, Praga Południe, Białołęka, Wawer & Praga Północ have zero plugs despite projecting 9,300+ EVs. Immediate greenfield investment required.

02

HIGH IMPACT · Revenue

Expand High-Income Zones to Unlock Revenue

SQL Q9 shows High tier pays zł142/session. PBI Page 4 confirms only 13 stations serve these areas. Adding 100 plugs in Śródmieście/Żoliborz could generate zł3M+ annually.

03

ML MODEL 1 · Planning

Use ML Forecasting for Plug Expansion Planning

Demand Forecasting model (R^2=0.87) predicts session volume by hour/day/district. Use predictions to pre-position plugs before EV registrations spike, not reactively after.

04

ML MODEL 2 · Revenue

Implement Dynamic Pricing Based on Regression Model

Cost Regression (R^2=0.91) identifies kWh as the #1 cost driver. Pair with hour-of-day feature to implement time-of-use pricing during 8 AM–3 PM peak window.

05

ML MODEL 3 · Marketing

Launch Segment-Based Customer Programs

Clustering reveals 3 clear segments. Target Cluster 1 (4,200 commuters) with monthly subscriptions, Cluster 0 (1,400 high-value) with fast-charge premium plans.

Thank You

Presented by

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14 Queries	4 Pages	3 Models	114K	65	zł11.4M
SQL	Power BI	ML Models	Sessions	Stations	Revenue